

NLP LAB: CONTEXT-FREE GRAMMARS (CFGs) USING THE PENN TREEBANK

1. Learning Outcomes

- Understand Penn Treebank phrase-structure annotations.
- Extract CFG rules from Treebank trees.
- Build probabilistic and non-probabilistic grammars.
- Parse sentences using induced PCFGs.
- Analyze rule frequencies, sparsity, and ambiguity.
- Explore syntactic variation in natural language treebanks.

2. Theory Overview

The Penn Treebank encodes syntactic structures using bracketed phrase-structure grammar.

CFGs consist of terminals, non-terminals, and production rules.

Treebanks allow extraction of grammar rules directly from annotated data.

PCFGs (Probabilistic CFGs) attach probabilities to grammar rules based on frequency.

3. LAB TASKS

Task 1 – Load and inspect Treebank trees

Use `nltk.corpus.treebank.parsed_sents()` to view sample trees.

Task 2 – Extract productions

Iterate over parsed sentences and collect `tree.productions()`.

Task 3 – Build a PCFG

Use `nltk.induce_pcfg()` on extracted productions.

Task 4 – Analyze rule frequencies

Compute most common and rare CFG rules.

Task 5 – Parse a sentence

Use ViterbiParser with the PCFG to parse:

"The dog chased the cat"

Task 6 – Ambiguity

Parse:

"The man saw the boy with the telescope"

Discuss PP-attachment ambiguity.

Task 7 – Sub-grammar

Extract only trees containing verbs and build a reduced grammar.

Task 8 – Visualize rule distribution

Plot histogram of rule frequencies.

Task 9 – Generate parse trees

Use pretty_print() to display trees.

Task 10 – Mini Project

A. Interactive tree viewer

B. Compare PCFG vs chart parser

C. Build a grammar of short sentences

4. Submission Requirements

- Notebook with all tasks
- Extracted grammar rules
- Parse trees for all examples
- Rule frequency analysis
- Discussion on ambiguity and grammar sparsity